

Design and Simulation of a Strongly Connected Distributed Network Using GNS3

Introduction

A distributed network allows devices to communicate and share resources. In a full mesh topology, every device is directly connected to every other device, providing high reliability and fault tolerance.

Mesh Topology

A full mesh topology is a network arrangement in which every device has a dedicated point-to-point connection with every other device. This creates multiple communication paths and eliminates a single point of failure.

Advantages

- High reliability
- Excellent fault tolerance
- Fast communication
- Easy fault detection
- Secure dedicated links

Disadvantages

- Expensive to install
- Large number of cables required
- Difficult to manage in large networks
- High installation complexity

Calculation

Formula: $\text{Connections} = n(n-1)/2$

For $n = 11$:

$\text{Connections} = 11 \times 10 / 2 = 55$.

Answer: A full mesh network with 11 devices requires 55 direct connections.

GNS3 Procedure

1. Create a new GNS3 project.
2. Add 11 routers or VPCS devices.

3. Label them R1 to R11.
4. Connect every device directly to every other device (55 links total).
5. Start all devices.
6. Assign IP addresses if required.
7. Test connectivity using ping.
8. Save the project.

Results

The topology was implemented as a full mesh with 11 devices and 55 links. Connectivity was verified using ping.

Conclusion

The assignment successfully demonstrated a strongly connected distributed network using a full mesh topology. The required number of connections for 11 devices is 55.